

PET and Gestational Hypertension 01: Language of Hypertensive Disease

Companion reading handout for simultaneous listening.

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This course is based on the PET and gestational hypertension study manual, ACOG Practice Bulletin two twenty two, NICE guideline NG133, the ISSHP international recommendations, the SOMANZ hypertension in pregnancy guideline, the Israeli position papers, the local Hadassah PET protocol, and Gabbe chapter thirty one on pre-eclampsia.

Opening Case

Let us begin with the sentence that causes most of the confusion.

Hypertension plus proteinuria equals pre-eclampsia.

That sentence is not completely wrong. It is just dangerously incomplete.

It was useful when the disease was taught in its older form, because proteinuria was the visible sign that hypertension had injured the kidney. But modern practice has moved beyond that. A patient can have pre-eclampsia without proteinuria. A patient can have gestational hypertension today and pre-eclampsia tomorrow. And a patient with severe-range blood pressure but no proteinuria is not safe just because the urine is quiet.

So the first job of this course is to replace a brittle definition with a layered clinical model.

Layer one is hypertension.

After twenty weeks, a previously normotensive patient develops a systolic blood pressure of at least one hundred forty, or a diastolic blood pressure of at least ninety. That is the doorway. It tells you that the hypertensive disorder conversation has started.

But it does not yet tell you the diagnosis.

It might be gestational hypertension. It might be pre-eclampsia. It might be chronic hypertension discovered late because no early pregnancy blood pressures were available. It might be white coat hypertension. It might be masked hypertension. Or it might be chronic hypertension with superimposed pre-eclampsia.

Layer two is organ involvement.

Now you ask, is this still isolated blood pressure, or has the disease reached an organ system?

If there is proteinuria, that is one form of organ involvement. But there are others. Platelets can fall. Creatinine can rise. Liver enzymes can rise. The patient can develop pulmonary edema. She can have persistent headache, visual symptoms, right upper quadrant pain, eclampsia, or HELLP syndrome. In ISSHP and SOMANZ-style definitions, the placenta and fetus can also show the disease through fetal growth restriction, abnormal Doppler, abruption, fetal death, or angiogenic imbalance.

That is the move from isolated hypertension to pre-eclampsia biology.

Layer three is severe features.

This is the urgency layer. Severe-range blood pressure, platelets below one hundred thousand, creatinine above one point one milligrams per deciliter or doubled from baseline, transaminases at least twice normal, pulmonary edema, persistent neurologic symptoms, or visual symptoms, all mean that the disease has crossed into a risk category where maternal deterioration can be rapid.

The practical point is essential.

Severe gestational hypertension is managed like pre-eclampsia with severe features.

Why? Because the immediate risk is not the name of the diagnosis. The immediate risk is stroke, seizure, pulmonary edema, renal injury, liver injury, abruption, and maternal collapse.

Layer four is gestational age.

Once you know the diagnosis and severity, the next question is whether time helps or harms. At thirty one weeks, stable disease may justify careful expectant management in the right center. At thirty seven weeks, continuing a non-severe hypertensive pregnancy usually buys little fetal benefit and keeps accumulating maternal risk. At thirty four weeks with severe features, most guidelines move toward delivery after stabilization.

So the diagnostic language is not a list. It is a sentence.

This is chronic hypertension, gestational hypertension, pre-eclampsia, or superimposed pre-eclampsia, at this gestational age, with or without severe features, with this maternal trajectory, and this fetal trajectory.

If you cannot say that sentence, you have not finished the assessment.

Now let us define the main terms.

Chronic hypertension is hypertension that was present before pregnancy, diagnosed before twenty weeks, or persists beyond the postpartum resolution window. ACOG

and the Hadassah protocol use persistence beyond twelve weeks postpartum as retrospective evidence of chronic hypertension. Chronic hypertension matters because it creates baseline noise. The patient may begin pregnancy with high BP, baseline proteinuria, or renal disease. So superimposed pre-eclampsia cannot be diagnosed by saying, her pressure is higher than before, and stopping there. You need new proteinuria, new organ dysfunction, or uteroplacental disease.

White coat hypertension means the clinic blood pressure is elevated, but home or ambulatory blood pressure is normal. ISSHP uses one hundred thirty five over eighty five as a typical out-of-office boundary. It is tempting to dismiss white coat hypertension as anxiety. Do not do that. It carries increased risk and needs home monitoring.

Masked hypertension is the opposite. The clinic pressure looks normal, but home or ambulatory readings are high. This becomes important when the patient has target organ findings or placental disease that do not match the clinic BP.

Gestational hypertension, or GHTN, is new hypertension after twenty weeks without proteinuria, without maternal end-organ dysfunction, and without fetal or placental features that your guideline system counts as pre-eclampsia.

The teaching point is that GHTN is often provisional.

It means, the hypertensive part of the syndrome has appeared, but we have not yet documented kidney, liver, platelet, lung, brain, or placental involvement. ACOG notes that up to half of women with GHTN eventually develop proteinuria or end-organ dysfunction, especially when hypertension begins before thirty two weeks. SOMANZ says that at first presentation, some women diagnosed as GHTN are actually developing pre-eclampsia but have not yet declared it.

So GHTN does not mean benign. It means not yet proven to be pre-eclampsia.

Pre-eclampsia, or PET, is new hypertension after twenty weeks plus proteinuria, maternal end-organ dysfunction, or, in broader international definitions, uteroplacental dysfunction.

This is where ACOG, ISSHP, NICE, SOMANZ, and local protocols differ slightly. ACOG does not treat fetal growth restriction alone as a severe feature. ISSHP and SOMANZ integrate uteroplacental dysfunction more directly into the diagnosis. But every source agrees on the clinically important point: fetal growth restriction, abnormal Doppler, abruption, and fetal compromise are not background noise. They are placental disease and must change surveillance and delivery thinking.

Pre-eclampsia with severe features is not just "more protein." In fact, the amount of proteinuria is not a severity meter. Severe features are the findings that predict maternal danger: severe BP, platelet consumption, renal dysfunction, hepatic injury, pulmonary edema, neurologic symptoms, visual symptoms, HELLP syndrome, eclampsia, and major maternal or fetal deterioration.

HELLP syndrome deserves special respect. It means hemolysis, elevated liver enzymes, and low platelets. It can appear without classic hypertension or proteinuria. It can present postpartum. It can look like right upper quadrant pain, nausea, vomiting, malaise, and abnormal labs. If you think of it only as a subtype of pre-eclampsia, you may miss its differential diagnosis: TTP, HUS, acute fatty liver of pregnancy, hepatitis, gallbladder disease, and other microangiopathies.

Eclampsia is new generalized seizure or unexplained coma in pregnancy or postpartum, not explained by another neurologic disorder, in the setting of pre-eclampsia physiology. It can occur before the chart has a neat pre-eclampsia label. That is why neurologic symptoms in a hypertensive pregnant or postpartum patient are never casual symptoms.

Closing Synthesis

Let us close the episode by returning to the model.

Do not ask only, is there protein?

Ask, first, is there new hypertension? Second, is there organ or placental involvement? Third, are there severe features? Fourth, what does gestational age make worth doing today?

That is the language of the course.

Resident Reflex

The resident reflex is this: GHTN is hypertension before organ disease is documented. PET is hypertension with maternal, placental, or fetal end-organ disease. Severe features and gestational age drive urgency.